

Sports Injury Risk Detection from Video

1. Title: Sports Injury Risk Detection from Video

2. Objective

Build an AI-powered Sports Injury Risk Detection Platform that analyzes athlete movement videos to identify biomechanical issues, detect abnormal movement patterns, assess injury risk factors, and predict potential injuries before they occur.

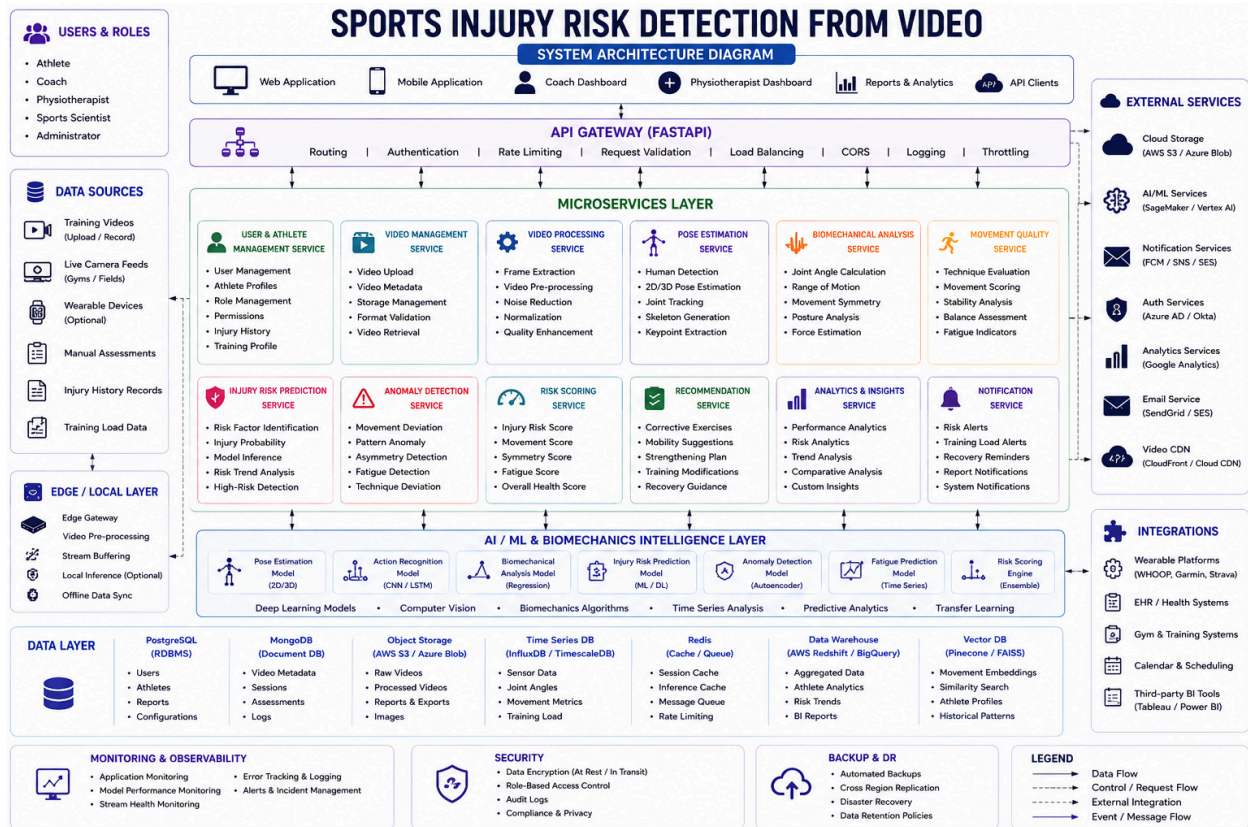
The platform leverages computer vision, pose estimation, biomechanics analysis, movement intelligence, and predictive analytics to help athletes, coaches, sports scientists, physiotherapists, and medical teams improve performance while reducing injury risks.

The solution enables proactive injury prevention through continuous movement monitoring, risk assessment, corrective recommendations, and performance analytics.

Outcomes

- Developed and deployed an AI-powered sports injury intelligence platform.
 - Implemented secure authentication and role-based access control.
 - Built athlete movement analysis and pose estimation systems.
 - Developed biomechanical assessment and injury risk prediction models.
 - Implemented movement anomaly detection and performance analytics.
 - Built corrective exercise recommendation workflows.
 - Developed dashboards for coaches, physiotherapists, and sports scientists.
 - Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.
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3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- User registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management

Roles

- Athlete

- Coach
- Physiotherapist
- Sports Scientist
- Administrator

2. Athlete Profile Management

- Athlete registration
- Injury history management
- Training profile management
- Performance tracking
- Physical assessment records

Athlete Information

- Athlete ID
- Sport Type
- Position
- Age
- Height
- Weight
- Injury History
- Training Load

3. Video Upload & Processing Engine

- Video upload
- Video preprocessing
- Frame extraction
- Motion enhancement
- Video quality validation

Supported Activities

- Running
- Sprinting
- Jumping
- Squatting
- Landing
- Throwing
- Cutting Movements
- Sport-Specific Drills

4. Pose Estimation Engine

- Human pose detection
- Joint tracking
- Skeleton generation
- Motion trajectory analysis
- Keypoint extraction

Key Body Points

- Head
- Shoulder
- Elbow
- Wrist
- Hip
- Knee
- Ankle
- Foot

5. Biomechanical Analysis Engine

- Joint angle analysis
- Range of motion analysis
- Movement symmetry evaluation
- Force estimation
- Posture assessment

Biomechanical Metrics

- Knee Valgus
- Hip Stability
- Trunk Lean
- Landing Mechanics
- Stride Length
- Joint Alignment
- Balance Metrics

6. Injury Risk Prediction Engine

- Risk factor identification
- Injury probability prediction
- Risk trend monitoring
- High-risk movement detection
- Personalized injury assessment

Injury Categories

- ACL Injury Risk
- Hamstring Injury Risk
- Ankle Sprain Risk
- Shoulder Injury Risk
- Lower Back Injury Risk

- Overuse Injury Risk

7. Movement Anomaly Detection Engine

- Movement deviation detection
- Technique assessment
- Motion inconsistency analysis
- Fatigue-related movement monitoring
- Performance decline detection

8. Risk Scoring Engine

- Injury risk scoring
- Movement quality scoring
- Biomechanical efficiency scoring
- Fatigue risk scoring
- Overall athlete health score

Weighted Scoring Model

Injury Risk Score =

- Biomechanical Deviations (35%)
- Historical Injury Factors (20%)
- Movement Asymmetry (20%)
- Training Load Indicators (15%)
- Fatigue Indicators (10%)

Risk Categories

- Low Risk
- Moderate Risk
- High Risk

- Critical Risk

9. Corrective Recommendation Engine

- Exercise recommendations
- Mobility improvement suggestions
- Strengthening recommendations
- Recovery planning
- Training modification suggestions

10. Dashboard & Analytics

Athlete Dashboard

- Injury risk score
- Movement analysis reports
- Progress tracking
- Exercise recommendations
- Performance trends

Coach Dashboard

- Team risk overview
- Athlete performance analytics
- Movement quality reports
- Training recommendations

Physiotherapist Dashboard

- Rehabilitation tracking
- Injury risk monitoring
- Movement correction analytics
- Recovery reports

Sports Scientist Dashboard

- Biomechanical analytics
- Team performance trends
- Injury prediction insights
- Research reports

Admin Dashboard

- User management
- Platform analytics
- System monitoring
- Report management

11. Notification & Alert System

- High-risk movement alerts
- Injury risk notifications
- Training load warnings
- Recovery reminders
- Assessment completion alerts

12. Reports & Export System

- Injury risk reports
- Biomechanical assessment reports
- Movement analysis reports
- Athlete performance reports
- Rehabilitation reports
- PDF export
- Excel export

13. Final Integration, Testing & Deployment

- Frontend and backend integration
 - API validation and testing
 - End-to-end workflow testing
 - Security testing
 - Performance optimization
 - Docker containerization
 - Production deployment
 - Monitoring and logging setup
 - Documentation and user guides
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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- Define project objectives and injury detection workflows.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Build athlete profile management workflows.
- Collect sports biomechanics datasets.

Recommended Datasets

Human3.6M Dataset

Purpose:

- Human pose estimation
- Joint tracking
- Movement analysis

MPII Human Pose Dataset

Purpose:

- Body keypoint detection
- Activity recognition

COCO Keypoints Dataset

Purpose:

- Pose estimation training
- Human motion analysis

SportsPose Dataset

Purpose:

- Sports-specific movement analysis
- Athlete posture assessment

FIFA Injury Dataset (Reference)

Purpose:

- Injury trend analysis
- Risk factor modeling

Outcomes

- Understand sports injury prediction workflows.
- Learn biomechanics and movement analytics concepts.
- Build frontend and backend project initialization.

- Working authentication and athlete management system.

Milestone 2: Week 3 & 4 — Pose Estimation & Biomechanical Analysis

Tasks

- Implement pose estimation engine.
- Build skeleton tracking workflows.
- Develop biomechanical analysis modules.
- Create movement quality assessment.
- Generate biomechanics reports.

Outcomes

- Pose estimation engine operational.
- Biomechanical analysis workflows functional.
- Movement assessment system completed.

Milestone 3: Week 5 & 6 — Injury Prediction & Recommendations

Tasks

- Implement injury risk prediction engine.
- Build movement anomaly detection workflows.
- Develop risk scoring models.
- Generate corrective recommendations.
- Create athlete intelligence dashboards.

Outcomes

- Injury prediction engine operational.
- Risk scoring system functional.
- Recommendation workflows completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build executive dashboards.
- Add reports and visualization modules.
- Implement testing and validations.
- Deploy platform using Docker and cloud services.
- Prepare final documentation and presentation.

Outcomes

- Fully deployed production-ready platform.
 - Sports injury intelligence systems operational.
 - Complete end-to-end injury prevention workflow demonstrable.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Project initialization completed.
- Authentication implemented.
- Athlete profile management operational.
- Pose datasets integrated.

Milestone 2 (Week 4)

- Pose estimation engine operational.
- Biomechanical analysis workflows functional.
- Movement quality assessment implemented.

Milestone 3 (Week 6)

- Injury prediction engine operational.

- Risk scoring functional.
- Recommendation workflows completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Dashboards and reporting systems operational.
 - End-to-end injury prediction workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend

- Python
- FastAPI

Frontend

- JavaScript
- React.js

Database

Primary Database

- PostgreSQL

Secondary Database

- MongoDB

AI & Machine Learning

Computer Vision

- OpenCV

- YOLOv8
- MediaPipe
- TensorFlow
- PyTorch

Biomechanics & Analytics

- Scikit-learn
- XGBoost
- Pandas
- NumPy

Pose Estimation Frameworks

- OpenPose
- MediaPipe Pose
- MoveNet
- Detectron2

Video Processing

- OpenCV
- FFmpeg
- DeepSORT

Visualization & Analytics

- Plotly
- Matplotlib
- Chart.js

Cloud & DevOps

- Docker

- AWS / Azure

Libraries & Frameworks

- FastAPI
- React.js
- Next.js
- Tailwind CSS
- JWT Authentication

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - GitHub Actions
 - Postman
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8. Performance Metrics

Pose Estimation Metrics

- Keypoint detection accuracy
- Pose tracking accuracy
- Joint localization accuracy

Biomechanical Analysis Metrics

- Joint angle estimation accuracy
- Symmetry analysis accuracy
- Movement quality assessment accuracy

Injury Prediction Metrics

- Injury risk prediction accuracy
- Risk classification precision
- False positive rate
- Early warning effectiveness

Recommendation Metrics

- Recommendation relevance
- Corrective exercise effectiveness
- Recovery improvement rate

System Performance Metrics

- Video processing latency
 - API response time
 - Dashboard loading speed
 - Concurrent video processing capacity
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9. Example Quantitative Goals

Injury Prevention

Identify biomechanical risks before injuries occur.

Movement Intelligence

Provide accurate movement quality and posture assessments.

Athlete Performance

Improve athletic performance through movement optimization.

Risk Prediction

Deliver early warnings for potential injury risks.

Platform Performance

Support large-scale sports video analytics and injury intelligence workflows while maintaining stable performance and responsiveness.